1. At absolute zero temperature (0 Kelvin), pure intrinsic silicon behaves as?
   a. good conductor
   b. insulator
   c. superconductor
   d. n-type semiconductor

   at 0 Kelvin electrons do not have sufficient thermal energy to move from valence band to conduction band. since since there are no free charge carriers for conduction it behaves as insulator

2. The majority carriers in p-type semiconductor are
   a. holes
   b. free electrons
   c. positive ions
   d. negative ions

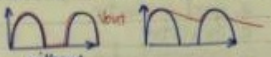
   p-type semiconductor is form by ~~Boron~~ doping silicon with boron group that has 3 valence electron while silicon has 4 valence electron when boron and silicon bonds together its lack 1 valence electron creating a hole, hole can be filled by electron from neighbouring bond, leaving another hole behind

3. A pn junction diode is forward biased when?
   a. the anode is more negative than cathode
   b. both terminal are at the same potential
   c. cathode is more positive than anode
   d. the p side (anode) is at a higher potential than n side

   ↳ reduce potential barrier and narrow depletion region at PN junction. The applied frib oppose the built in electric field allow majority carriers cross the junction.

4. Using the constant voltage drop model, the forward voltage of a conducting silicon diode is about?
   a. 0.1 V
   b. 0.3 V
   c. 0.7 V
   d. 1.5 V

5. The main purpose of the capacitor placed across the load of a rectifier is
   a. increase the ripple voltage
   b. raise the ripple frequency
   c. protect the diode from reverse voltage
   d. smooth the output by reducing ripple



↳ without filter, rectifier output falls to zero between peak
capacitor store charger during voltage peaks and release it to load
between peaks. that reduce ripple and smooth the output

6. For NPN BJT operating in active ~~not~~ mode
   a. base-emitter junction forward biased, base-collector junction reversed bia
   b. both juction frib
   c. n ——— n frib rvb
   d. n ——— n reversed biased, n ——— n frib

7. In NPN BJT, the most heavily doped region is
   a. base
   b. emitter — cause it need to inject charge carriers to base region
   c. collector — base region is thinly dope to allow carriers pass through
   d. all regions are doped equally collector, to blokt carrier ≠ riot in flow I

8. Which relationship among ~~of~~ bjt terminal currents is correct?
   a. $I_C = I_E + I_B$
   b. $I_B = I_C + I_E$ using KCL
   c. $I_E = I_C - I_B$
   d. $I_E = I_B + I_C$

9. When an NPN bjt is used on closed switch, it operates in the
   a. sat mode
   b. active mode
   c. cut off mode
   d. reverse active mode

10. Connecting a smaller load resistance R_L to the output makes the gain magnitude?

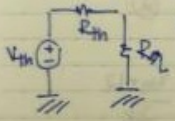
a. decrease
b. increase
c. stay the same
d. change sign

for common emitter $A_v = -g_m(R_C \| R_L)$
a parallel combination is smaller than either resistor

example $R_C = 6.8\,k\Omega \rightarrow R_L = 10k\Omega \quad R_C \| R_L = 4.05k\Omega$

$R_L = 1k\Omega \quad R_C \| R_L = 0.87k\Omega$

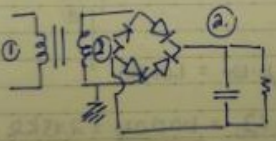
Thevenin common



$A_v = \frac{V_{th} \cdot R_L}{R_L + R_0} / V_{th} = R_L / R_L + R_0$

as R_L ↓ Gain ↓ also!

1. input = 100 V_{rms} 50 Hz output $V_{DC} = 15V$ $V_r = 0.5V$
using full wave bridge $R_L = 300\,\Omega$



ans $V_0 = V_m e^{-t/RC}$

into $-t/RC$ notation binary $V_0 = V_m(1 - \frac{T}{RC})$

$V_{ripple} = V_m \cdot \frac{T}{RC}$

$0.5 = V_m \cdot \frac{1/60}{300 \cdot C}$

(b) $\frac{N_1}{N_2} = \frac{V_1}{V_2} \quad \frac{N_1}{N_2} = \frac{100\sqrt{2}}{15.25} = \frac{9.27}{1}$

(d) ripple freq = ~~50 Hz~~ 100 Hz

into $V_m = V_{DC} + \frac{V_r}{2}$

$= 15 + 0.5/2 = 15.25$

~~$P_{IV} = 2V_p = 2(15.25) = 30.5V$~~

$P_{IV} = V_p = 15.25\,V$

for $C = 1mF$ (c)

Part 2 — Problems (20 points)

Problem 1: Rectifier Design (10 points)

Design a power supply that converts the 100 V_{rms}, 50 Hz mains to a dc output voltage V_{DC} = 15 V with a peak-to-peak ripple voltage V_r = 0.5 V. Use a step-down transformer, a **full-wave bridge rectifier**, and a capacitor filter feeding a load R_L = 300 Ω. Assume **ideal diodes** (zero forward drop) and an ideal transformer.

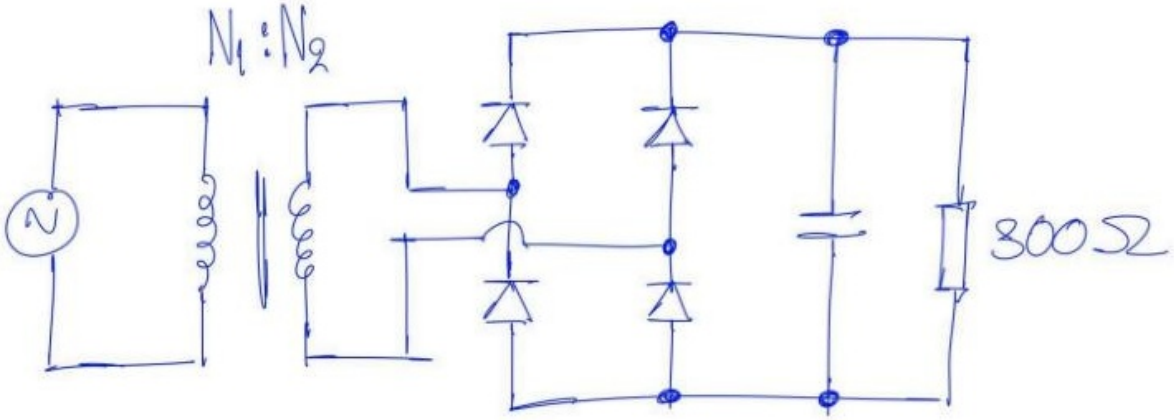
Take the dc output as the average capacitor voltage, $V_{DC} \approx V_p - V_r/2$. Note that $\frac{V_r}{V_p} = \frac{1}{f_r R_L C}$

(a) Draw the complete circuit, labeling the transformer, the four diodes, C, and R_L. [2 pts]

(b) Find the required peak output voltage V_p and the transformer turns ratio N_1:N_2. [3 pts]

(c) Find the required filter capacitance C. [3 pts]

(d) Find the peak inverse voltage (PIV) rating of each diode and the ripple frequency. [2 pts]



V_{DC} @ Load = 15V

from $V_{DC} \approx V_p - V_r/2$

$V_p \approx V_{DC} + V_r/2$

$\approx 15 + 0.25$

≈ 15.25

$V_r = \frac{V_m}{fRC}$

$C = \frac{V_m}{V_r f R} = \frac{15.25}{0.5 \times 100 \times 300}$

$C = 1.016$ mF

PIV = 15.25V

$f_r = 2f_s = 100$ Hz

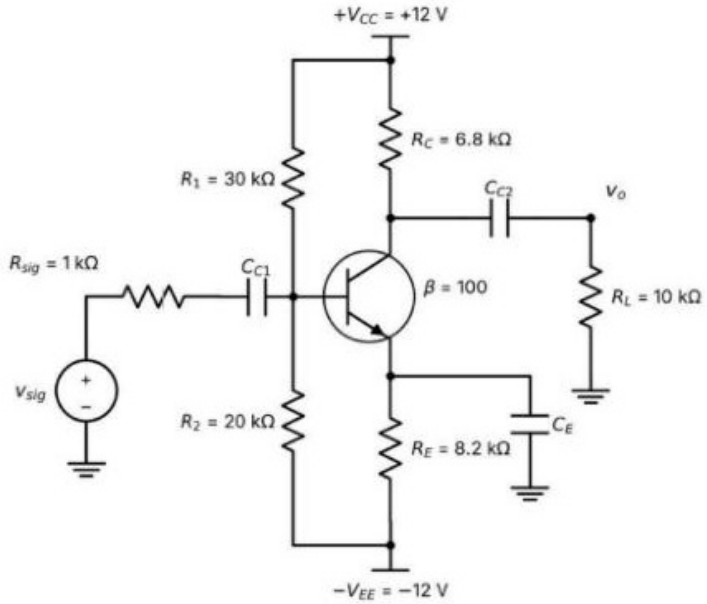
$\frac{100\sqrt{2}}{15.25} = \frac{N_1}{N_2}$

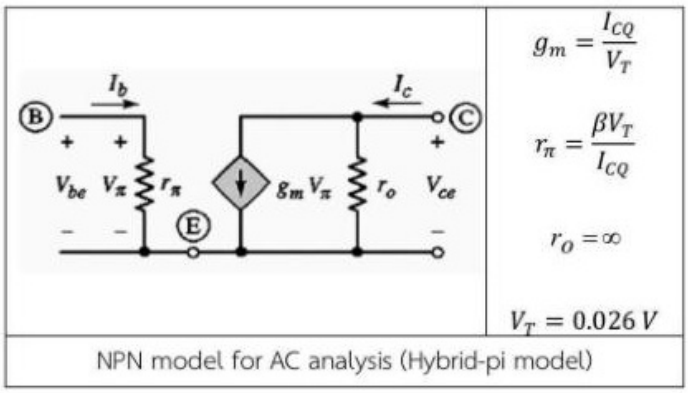
$9.27 = N_1 : N_2$

Student ID: _____

Problem 2: Fully-Bypassed Common-Emitter Amplifier (10 points)

The NPN amplifier below is biased with a dual (±12 V) supply; the emitter resistor R_E is fully bypassed by C_E. Assume ß(beta) = 100 and $V_{BE(on)}$ = 0.7 V.





NPN model for AC analysis (Hybrid-pi model)

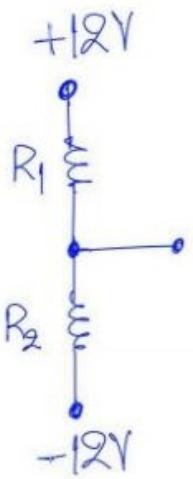
(a) DC analysis: find I_B, I_C, and V_{CE}, and verify the transistor is in the active mode. [4 pts]

(b) Find the small-signal parameters g_m and r_π. [2 pts]

(c) Draw the small-signal (hybrid-π) equivalent circuit of the amplifier. [2 pts]

(d) Find the overall gain $A_v = v_o/v_{sig}$. [2 pts]

DC



$$\frac{12-V_B}{R_1} = \frac{V_B+12}{R_2}$$

$$12R_2 - V_BR_2 = V_BR_1 + 12R_1$$

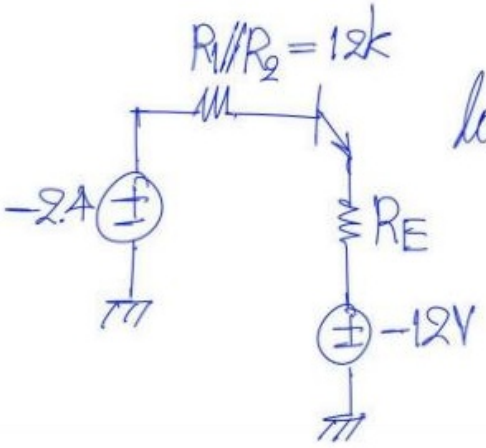
$$12(R_2-R_1) = V_B(R_1+R_2)$$

$$V_B = \frac{12(R_2-R_1)}{R_1+R_2}$$

$$V_B = -2.4\,V$$

$$V_B = V_\circ \cdot \frac{R_2}{R_1+R_2} = 24 \cdot \frac{20k}{50k} = 9.6\,V$$

$$V_B - 12V = 9.6 - 12 = -2.4\,V$$



loop; $2.4 + I_B(12k) + 0.7 + (\beta+1)I_B\,8.2k - 12 = 0$

$$12k\,I_B + 101 \cdot I_B \cdot 8.2k = 12 - 3.1$$

$$I_B(12k + 101 \times 8.2k) = 8.9$$

$$I_B = 1.059 \times 10^{-5} = 10.59\,\mu A$$

$$I_C = 1.059\,mA$$

$$I_E = 1.069\,mA$$

Problem 2 — work space (continued)

$I_C = \frac{12 - V_C}{R_C} \longrightarrow V_C = 12 - I_C R_C = 12 - 1.059m \times 6.8k$

$= 4.7988$

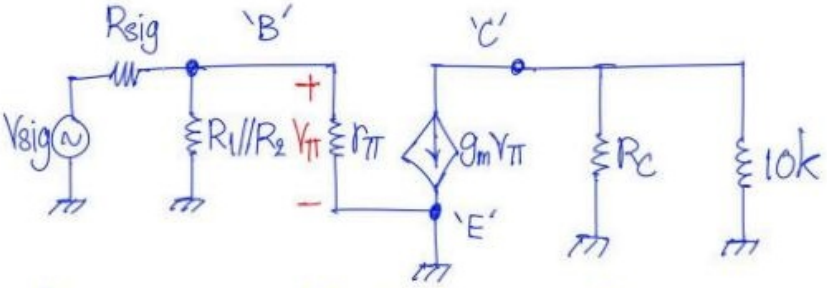
$I_E = \frac{V_E - (-12)}{R_E}$ $\quad V_E = -12 + I_E R_E = -12 + 1.069m \times 8.2k$

$= -3.2342$

$V_{CE} = 4.7988 - (-3.2342)$

$V_{CE} = 8.033 > V_{CE(sat)} \equiv$ Active mode

AC



$g_m = \frac{I_C}{V_T} = \frac{1.059m}{26m} = 40.73m$

$r_\pi = \frac{\beta V_T}{I_C} = 2455.1$

$V_o = -g_m V_\pi \cdot (R_C \| R_L)$ ——①

$V_\pi = V_{sig} \cdot \frac{(R_1 \| R_2 \| r_\pi)}{R_{sig} + (R_1 \| R_2 \| r_\pi)}$ ——②

②in①; $V_o = -g_m \frac{V_{sig}(R_1 \| R_2 \| r_\pi) \cdot (R_C \| R_L)}{R_{sig} + (R_1 \| R_2 \| r_\pi)}$

$\frac{V_o}{V_{sig}} = -g_m \frac{(R_1 \| R_2 \| r_\pi)}{R_{sig} + (R_1 \| R_2 \| r_\pi)} \cdot (R_C \| R_L)$